Nick Choksi

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Appointments

2025-Present **51 Pegasi b Postdoctoral Fellow**, *Caltech*.
Host: Konstantin Batygin

Education

2019-2025 **Ph.D. in Astrophysics**, *University of California*, *Berkeley*.
Advisor: Eugene Chiang

2019 **B.A. in Physics & Astrophysics**, *University of California*, *Berkeley*,
Advisor: Oleg Gnedin, Marta Volonteri, Peter Behroozi.

Research interests

theoretical astrophysics, planet formation, gravitational dynamics, circumstellar accretion disks, stellar cluster formation

Publications

14 lead-authored publications with 428 citations and $h = 10$
18 total publications with 855 citations and $h = 14$

1. **Choksi**, Gao, Xuan, Balmer, & Mang, "The accretionally heated atmospheres of protoplanets and brown dwarfs," arXiv 2610.12562.
2. **Choksi**, Lithwick, Chiang, & Li, "Two-stage disruption of resonant chains," MNRAS in press, arXiv 2604.05035.
3. **Choksi** & Chiang, "Spectral Energy Distributions of Disc-Embedded Accreting Protoplanets," MNRAS 537 2945 (2025), arXiv 2403.10057.
4. Dai, Goldberg, Batygin, van Saders, Chiang, **Choksi**, et al., "The Prevalence of Resonance Among Young, Close-in Planets," ApJ 168, 239 (2024), arXiv 2406.06885.
5. Li, Chiang, **Choksi**, and Dai, "The Resonant Remains of Broken Chains from Major and Minor Mergers," ApJ in press, arXiv 2408.10206.
6. **Choksi**, Chiang, Fung, & Zhu, "The maximum accretion rate of a protoplanet: how fast can runaway be?," MNRAS 525, 2806 (2023), arXiv 2305.01684.
7. **Choksi** & Chiang, "Exciting the TTV Phases of Resonant Sub-Neptunes," MNRAS 522, 1914 (2023), arXiv 2211.15701.
8. Rein & **Choksi**, "An Implementation of Stochastic Forces for the N-body Code `REBOUND`," RNAAS 6, 5 (2022), arXiv 2205.06757.
9. **Choksi** & Chiang, "Testing planet formation from the ultraviolet to the millimeter," MNRAS 510, 1657 (2021), arXiv 2110.00029.
10. **Choksi**, Chiang, Connolly, Gainsforth, and Westphal, "Chondrules from high-velocity collisions: thermal histories and the agglomeration problem," MNRAS 503, 3297 (2021),

arXiv 2009.10093.

11. **Choksi** & Chiang, "Sub-Neptune Formation: The View from Resonant Planets," MNRAS 495, 4192 (2020), arXiv 2003.03388.
12. **Choksi** & Kruijssen, "On the initial mass-radius relation of stellar clusters," MNRAS 507, 5492, arXiv 1912.05560.
13. **Choksi** & Gnedin, "Origins of scaling relations of globular cluster systems," MNRAS 488, 5409 (2019), arXiv 1905.05199.
14. **Choksi** & Gnedin, "Formation of Globular Cluster Systems II: Impact of the cutoff of the cluster initial mass function," MNRAS 486, 331 (2019), arXiv 1810.01888.
15. **Choksi**, Volonteri, Colpi, Gnedin, and Li, "The star clusters that make black hole binaries across cosmic time," ApJ 873, 100 (2019), arXiv 1809.01164.
16. El-Badry, Quataert, Weisz, **Choksi**, and Boylan-Kolchin, "The formation and hierarchical assembly of globular cluster populations," MNRAS 482, 4528 (2018), arXiv 1805.03652.
17. **Choksi**, Gnedin, and Li, "Formation of globular cluster systems: from dwarf galaxies to giants," MNRAS 480, 2343 (2018), arXiv 1801.03515.
18. **Choksi**, Behroozi, Volonteri, Schneider, Ma, Silk, and Moster, "Recoiling supermassive black hole escape velocities from dark matter halos," MNRAS 472, 1526 (2017), arXiv 1707.06220.

PI/co-PI telescope proposals

1. *Measuring the composition and circumplanetary environment of a gap-clearing protoplanet,* James Webb Space Telescope Cycle 5 GO 9993, 44 hours.

Invited Seminars and Conference Talks

UCLA	Colloquium, 2026
Los Alamos National Lab	Distinguished Seminar Series, 2026
Shanghai	Invited Conference Talk, 2025
International Conference on Exoplanets and Planet Formation	
CIERA	Theory Seminar, 2025
Princeton	Exoplanet Seminar, 2024
Caltech	TAPIR Seminar, 2024
McGill	Trottier Space Sciences Seminar, 2023
Hawaii IfA	Star and Planet Formation Seminar, 2023
Harvard	CfA Seminar, 2023
Lorentz Center	Invited Conference Talk, 2019
Formation of stars & massive clusters in dwarf galaxies across cosmic time	

Honors & Awards

2025 Mary Elizabeth Uhl Award, Berkeley Astronomy

2024 Robert J. Trumpler Award, Berkeley Astronomy

2024 UC Dissertation-Year Fellow ($40,000)

2019-2024 NSF Graduate Research Fellowship

2020 Esper Larsen Jr. Grant, Berkeley Earth & Planetary Science Department ($20,000)

2021 H2H8 Fellow ($10,000)

2019 Student commencement speaker, Berkeley Astronomy

2019 Finalist, Hertz Fellowship

2018 Isidore Pomerantz Award, Berkeley Physics

Service, Outreach, and Teaching

Current **Referee**, *MNRAS, ApJ, Icarus*.

2021 **Team member, Berkeley Discover Astronomy & Physics**. Collaborator on successful proposal ($800,000) to revamp undergraduate teaching in physics & astronomy at Berkeley.

5 semesters **Graduate student instructor**, *Astrophysics I & II, Astronomy for non-majors*.